

How to Use This Atlas

EACH DOUBLE-PAGE SPREAD in this atlas covers a particular type of habitat (the place where an animal lives). For example, the spread shown below is about European conifer forests. Habitats are arranged by continent, and there is a section in the book for each of

the continents—North America, South America, Europe, Africa, Asia, Australasia, and Antarctica. The heading at the top of each page tells you which section you are in. Below you can see what the maps and symbols on each spread show, and what the abbreviations stand for.

Where on Earth?
This globe shows you where in the world the habitat featured on the spread is situated and its rough extent. On this page, for example, the red color shows the area covered by the European conifer forests.

Wild cat
(*Felis silvestris*)

Latin names
Scientists have given each species of animal a Latin name, so that people all over the world can use the same name, no matter what language they speak. An animal's Latin name is divided into two parts. The first part is a group name given to a number of similar animals. For example, *Felis* is the group name given to all small cats. The second part of the name identifies the particular species of animal and often describes one of its specific characteristics. The full name for the wild cat shown here is *Felis silvestris*, which means "cat of the woods."

How big?
Labels next to each animal give the animal's vital statistics: its height, length, or wingspan. Like people, animals of the same species vary in size, so the measurement can only be approximate. Individual animals may be bigger or smaller than this size.

EUROPE

Conifer Forests

A THICK BAND OF DENSE EVERGREEN FOREST stretches across the northern parts of Europe, covering large areas of Scotland and Scandinavia. There are smaller evergreen forests farther south, such as the Black Forest in Germany and the Ardennes in Belgium. The most common trees in these forests are conifers (trees that have cones), such as pines, spruces, and firs. In recent years, acid rain, which is especially harmful to trees with needlelike leaves, has damaged many European conifer forests.

Animals that live in these forests have to survive in a severe climate. The winters are bitterly cold, but most conifer trees keep their leaves year round and provide some shelter. Some forest animals, such as the stoat, grow white coats in the winter so that they are camouflaged against the snow. Other animals, such as the wood ant, hibernate during the winter, while some birds, such as the osprey, migrate south to warmer places.

Long ears
The brown long-eared bat's huge ears are three-quarters the length of its body. They are so big that a young bat cannot hold its ears up straight until it is old enough to fly. The bat feeds on moths, midges, and flies. During the cold winter months, it hibernates in a cave.

Underground city
Wood ants build huge nests on the forest floor from pine needles and other plant material. Nests keep the ants warm in winter, when they hibernate in the soil beneath the mound. When threatened, a wood ant sprays its enemy with a stinging liquid called formic acid from glands on its abdomen.

Striped tail
The wild cat is closely related to the domestic cat, but it is slightly bigger and has a thicker tail with black rings on it. The wild cat hunts at night for small mammals, birds, and insects. The forests provide it with cover for hunting.

Crossed beak
The crossbill uses its crossed beak to tear open pine cones, so it can lick out the seeds. Adult crossbills regurgitate partly digested seeds to feed their young. Every few years, crossbills move out of their normal breeding areas and invade other parts of Europe. If conditions are good, they may settle in the new area for one or more seasons.

False ears
The "ears" of the northern long-eared owl are only tufts of feathers—in its ear openings are on the sides of its head. It hunts at night using its sharp eyesight and good hearing to find small mammals on the floor.

Fish snatcher
The osprey feeds on fish that it snatches from lakes. It has long, sharp claws and horny spines under its toes, which enable it to grip a slippery fish. An adult osprey can carry a fish weighing up to 4.5 lbs (2 kg). In the fall, the osprey migrates to Africa, where the weather is warmer and there are plenty of fish for it to eat.

Eaten alive
The female giant ichneumon lays her eggs next to the larvae of wood wasps, which the young grubs feed on when they hatch. Because wasp larvae tunnel deep inside tree trunks, the ichneumon drills through the wood with an egg-laying tube, which is 1.5 in (4 cm) long.

Antler fights
In the fall mating season, called the rut, the male red deer fights rival males with his antlers to win females for mating. He sheds his antlers each spring. New ones grow in time for the next mating season.

Flying acrobat
The pine marten is an acrobatic and swift nighttime hunter. It has strong legs, broad pads, and long claws that help it climb, and its bushy tail helps it balance. It eats many things, from small birds to rats, beetles, and fruit. It hunts on the ground and in the trees.

Dancing display
In spring, the male capercaillie puts on a display to attract females. He fans out his tail, points his neck upward, and makes odd gurgling sounds. He may even jump and clap his wings. Displaying capercaillies are very aggressive and will threaten deer, sheep, or even humans who disturb them.

Smelly protection
If it is threatened, the polecat produces a foul-smelling liquid from glands under its tail. The polecat also uses this scent to mark its territory. polecats do not hibernate, and hunt for small mammals all year round.

Wild cat
(*Felis silvestris*)
Body length: up to 28.5 in (74 cm)
Tail: up to 14.5 in (37 cm)

Red crossbill
(*Loxia curvirostris*)
Length: up to 7.8 in (20 cm)

Osprey (*Pandion haliaetus*)
Length: up to 28.1 in (64 cm)
Wingspan: up to 58.9 in (1.7 m)

Northern long-eared owl (*Asio otus*)
Length: up to 16 in (41 cm)
Wingspan: up to 38.5 in (1 m)

Wood ant (*Formica rufa*)
Length: up to 0.35 in (0.9 cm)

Western red deer (*Cervus elaphus*)
Height at shoulder: up to 48.3 in (1.3 m)
Body length: up to 88.9 in (2.3 m)

Polecat (*Mustela putorius*)
Body: up to 18 in (46 cm)

Giant ichneumon (*Ichneumon persusator*)
Length: up to 1.6 in (4 cm)

Western capercaillie (*Tetrao urogallus*)
Length: up to 38.9 in (1.15 m)

European pine marten (*Martes martes*)
Body length: up to 23 in (58 cm)

Brown long-eared bat (*Myotis myotis*)
Wingspan: up to 11 in (28 cm)

Wild cat (*Felis silvestris*)
Body length: up to 28.5 in (74 cm)
Tail: up to 14.5 in (37 cm)

Red crossbill (*Loxia curvirostris*)
Length: up to 7.8 in (20 cm)

Osprey (*Pandion haliaetus*)
Length: up to 28.1 in (64 cm)
Wingspan: up to 58.9 in (1.7 m)

Northern long-eared owl (*Asio otus*)
Length: up to 16 in (41 cm)
Wingspan: up to 38.5 in (1 m)

Wood ant (*Formica rufa*)
Length: up to 0.35 in (0.9 cm)

Western red deer (*Cervus elaphus*)
Height at shoulder: up to 48.3 in (1.3 m)
Body length: up to 88.9 in (2.3 m)

Polecat (*Mustela putorius*)
Body: up to 18 in (46 cm)

Giant ichneumon (*Ichneumon persusator*)
Length: up to 1.6 in (4 cm)

Western capercaillie (*Tetrao urogallus*)
Length: up to 38.9 in (1.15 m)

European pine marten (*Martes martes*)
Body length: up to 23 in (58 cm)

Brown long-eared bat (*Myotis myotis*)
Wingspan: up to 11 in (28 cm)

Winters in the forests are bitterly cold. With more covering the ground for up to half the year.

Many European conifer forests have been planted by people and are harvested for their timber.

28 29

ABBREVIATIONS USED IN THE BOOK			
mm	millimeter	sq mile	square mile
cm	centimeter	kph	kilometers per hour
in	inch	mph	miles per hour
m	meter	kg	kilogram
ft	foot	lb	pound
km	kilometer	°C	centigrade
sq km	square kilometer	°F	fahrenheit

Scale
You can use this scale to figure out the size of the area shown on the map. The maps in the book have been drawn to different scales.

Animal symbols
The animal symbols on the map show the main area where the animals can be found, but some animals are widely distributed over the whole region. There is one symbol for each of the animals illustrated on the spread.

Map
The map shows the area of the habitat featured on the spread and surrounding regions. On these pages, for example, the map shows a large part of Europe, covering the conifer forests and the areas around them. The map also shows major geographic features in the region, and where the animals live. You can see the shape and position of the forests themselves on the globe in the top-left corner of the page.

Photos
The photographs around the map show you what the habitat looks like and what sort of vegetation can be found there.